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Exp. No: 01 Date:	Install Virtual box with Linux Top of Windows

Aim

To Install Virtual box / VM ware Work station with different flavours of linux or windows OS on top of windows7 or 8.

Procedure

Steps to install VirtualBox

1. Download VirtualBox installer for windows.
2. The installer can be downloaded from the link:
3. https://www.virtualbox.org/wiki/Downloads
4. Click "Windows host" to download the binary version for windows host.
5. The installer file downloaded will have the file name format like "VirtualBox-VersionNumber-BuildNumber-Win.exe". Example: VirtualBox-6.1.12-139181-Win.exe.
6. Double click on the installer to launch the setup Wizard. Click on Next to continue.
7. Custom setup dialog box will be opened. Accept the default settings and click next.
8. Select the way you want the features to be installed. You can accept the default and click next.
9. A dialog box opens with Network Interfaces warning. Click Yes to proceed.
10. Click install to begin the installation process.
11. When prompted with a message to install (Trust) Oracle Universal Serial Bus, click Install to continue.
12. After the installation completes, click finish to exit the setup wizard.
13. Launch the Oracle VM VirtualBox.

Steps to create a virtual machine [Ubuntu] in VirtualBox

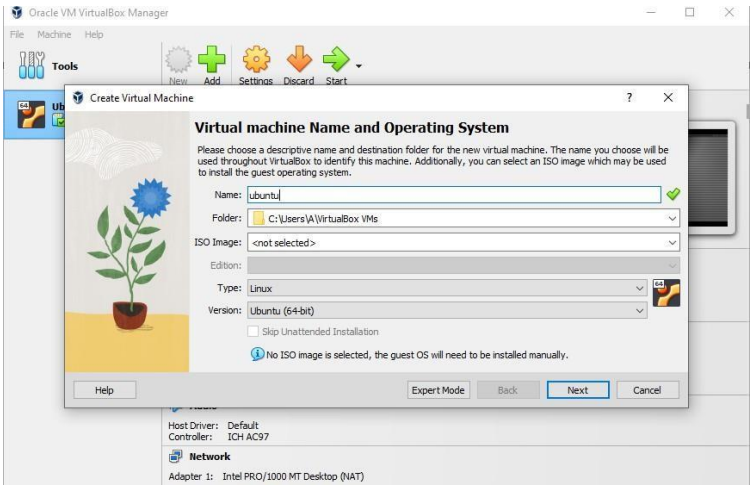
1. Open the Oracle VM VirtualBox.
2. Click New icon or "Ctrl + N" to create a new virtual machine.
3. Enter a name for the new virtual machine. Choose the Type and Version. Note that VirtualBox automatically changes 'Type' to Linux and 'Version' to 'Ubuntu (64 bit)' if the name is given as 'Ubuntu'. Click Next.
4. Select the amount of RAM to use. The ideal amount of RAM will automatically be selected. Do not increase the RAM into the red section of the slider; keep the slider in the green section.
5. Accept the default 'Create a virtual hard drive now' and click 'Create' button.
6. Choose the hard disk file type as VDI (VirtualBox Disk Image). Click Next.
7. Click Next to accept the default option 'Dynamically allocated' for storage on physical hard drive.

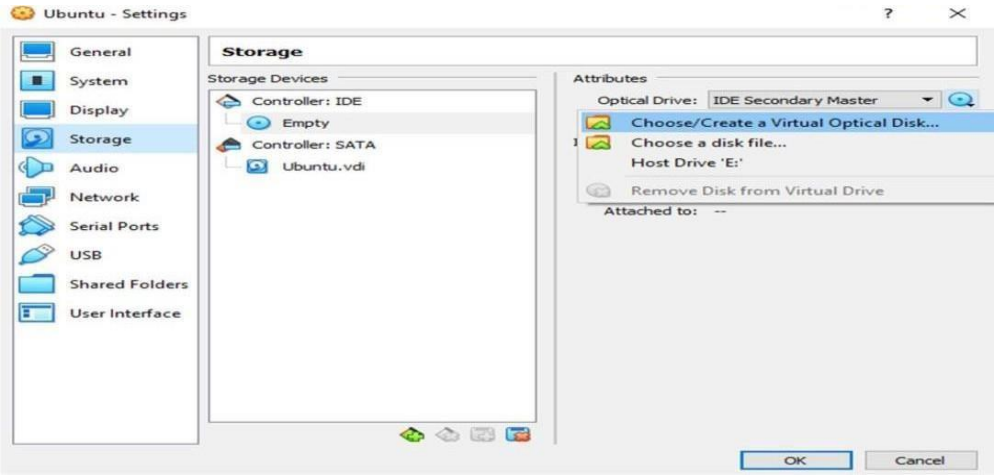
8. Select the size of the virtual hard disk and click create.
9. The newly created virtual machine will be displayed in the dashboard.
10. Download the ISO file [Ubuntu disk image file]. Latest version of Ubuntu iso file can be downloaded from the link https://ubuntu.com/download/desktop. Click Download button.
11. For previous versions, goto http://releases.ubuntu.com. Choose the preferred version of Ubuntu and download the iso file.
12. To setup the Ubuntu disk image file (iso file) goto settings.
13. Click Storage. Under “Storage Devices” section click “Empty”.
14. In Attributes section, click the disk image and then "Choose Virtual Optical Disk File".
15. Browse and select the downloaded iso file. Click ok.
16. Select the newly created virtual machine in the dashboard and click start button.
17. In the welcome screen, click “Install Ubuntu” button.
18. Click 'Continue' button.
19. Make sure 'Erase disk and install Ubuntu' option is selected and click 'Install Now' button.
20. Choose the default and click continue.
21. Setup up your profile by creating username and password.
22. After installation is complete, click 'Restart Now' button and follow the instructions.
23. The Ubuntu OS is ready to use. Login with the username and password.

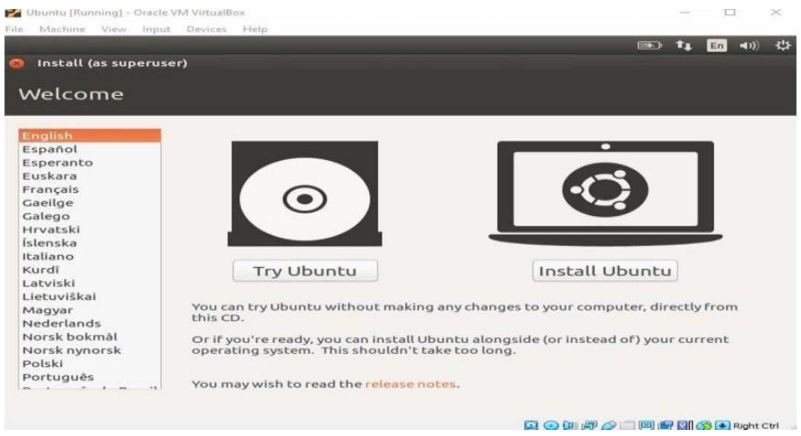
Input



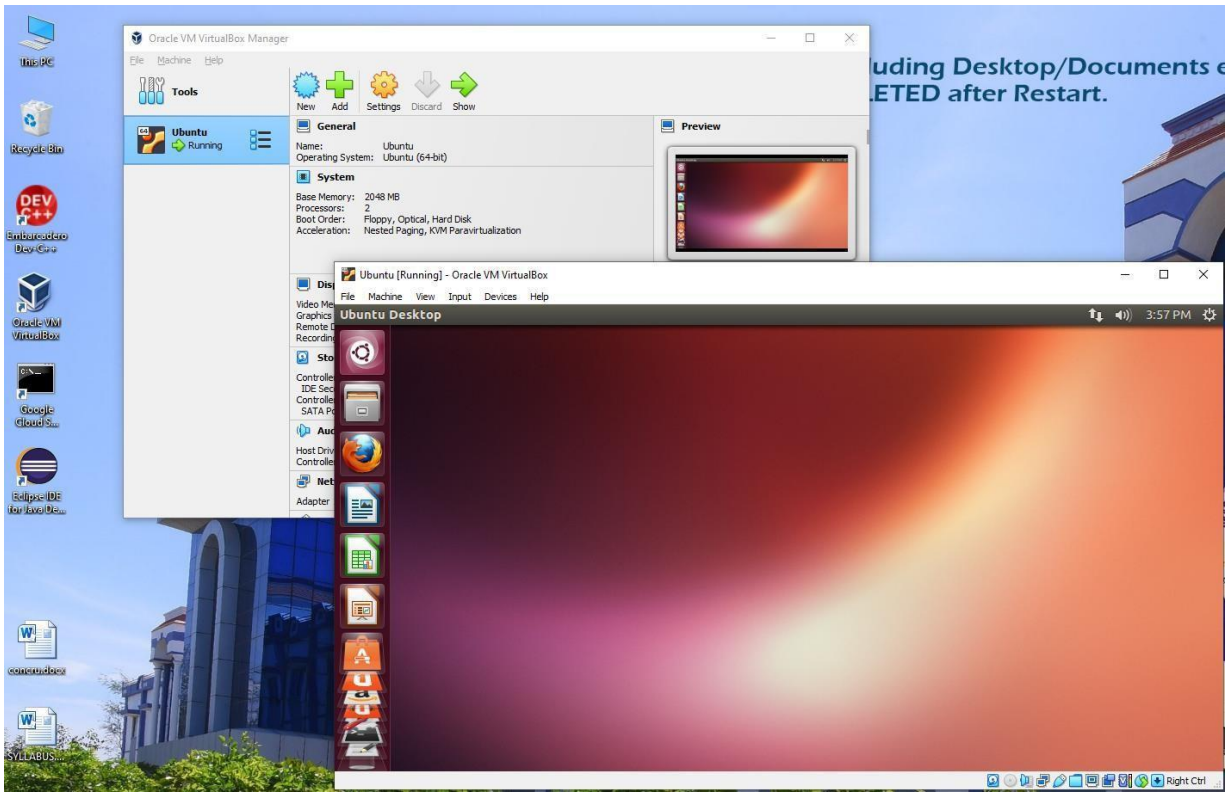



Oracle VM VirtualBox Manager
File Machine Help
Tools
New Add Settings Discard Start
Create Virtual Machine
Virtual machine Name and Operating System
Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.
Name: ubuntu
Folder: C:\Users\A\VirtualBox VMs
ISO Image: <not selected>
Edition:
Type: Linux
Version: Ubuntu (64-bit)
Skip Unattended Installation
No ISO image is selected, the guest OS will need to be installed manually.
Help
Expert Mode
Back
Next
Cancel
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)


Ubuntu - Settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Storage
Storage Devices
Controller: IDE
Empty
Controller: SATA
Ubuntu.vdi
Attributes
Optical Drive: IDE Secondary Master
Choose/Create a Virtual Optical Disk...
Choose a disk file...
Host Drive 'E:'
Remove Disk from Virtual Drive
Attached to: --
OK
Cancel


Ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Install (as superuser)
Welcome
English
Español
Esperanto
Euskara
Français
Gaeilge
Galego
Hrvatski
Íslenska
Italiano
Kurdî
Latviski
Lietuviškai
Magyar
Nederlands
Norsk bokmål
Norsk nynorsk
Polski
Português
Try Ubuntu
Install Ubuntu
You can try Ubuntu without making any changes to your computer, directly from this CD.
Or if you're ready, you can install Ubuntu alongside (or instead of) your current operating system. This shouldn't take too long.
You may wish to read the release notes.
Right Ctrl

Output



Result

Thus the procedure to run the virtual machine of different configuration.

Exp. No: 02 Date:	Install a C compiler in the virtual machine

Aim

To Install a C compiler in the virtual machine created using virtual box and execute Simple Programs.

Procedure

1. Launch the virtual box and open the virtual machine (Ubuntu).
2. Run the following command in the virtual machine terminal.

   $ sudo apt-get update

   $ sudo apt-get install gcc

   It will install all the necessary packages for gcc complier.
3. Type the C program in the text editor and save the file with .c extention.

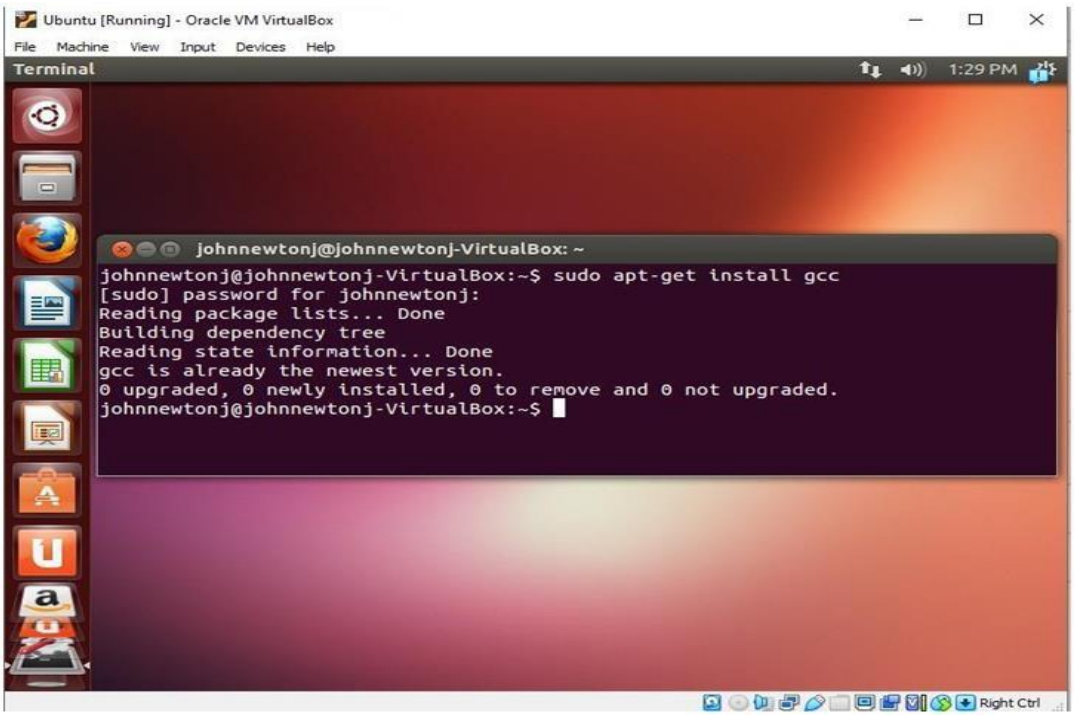
```
//fact.c
#include<stdio.h>
void main()
{       int n, fact=1;
        int i=1;
        printf("Enter a positive integer:");
        scanf("%d",&n);
        if(n==0)
        {       fact=1;         }
        else
        {       while(i<=n)
                {
                        fact *= i ;
                        i++;
                }
                printf("The factorial of %d is %d ",n,fact);
        }
}
```

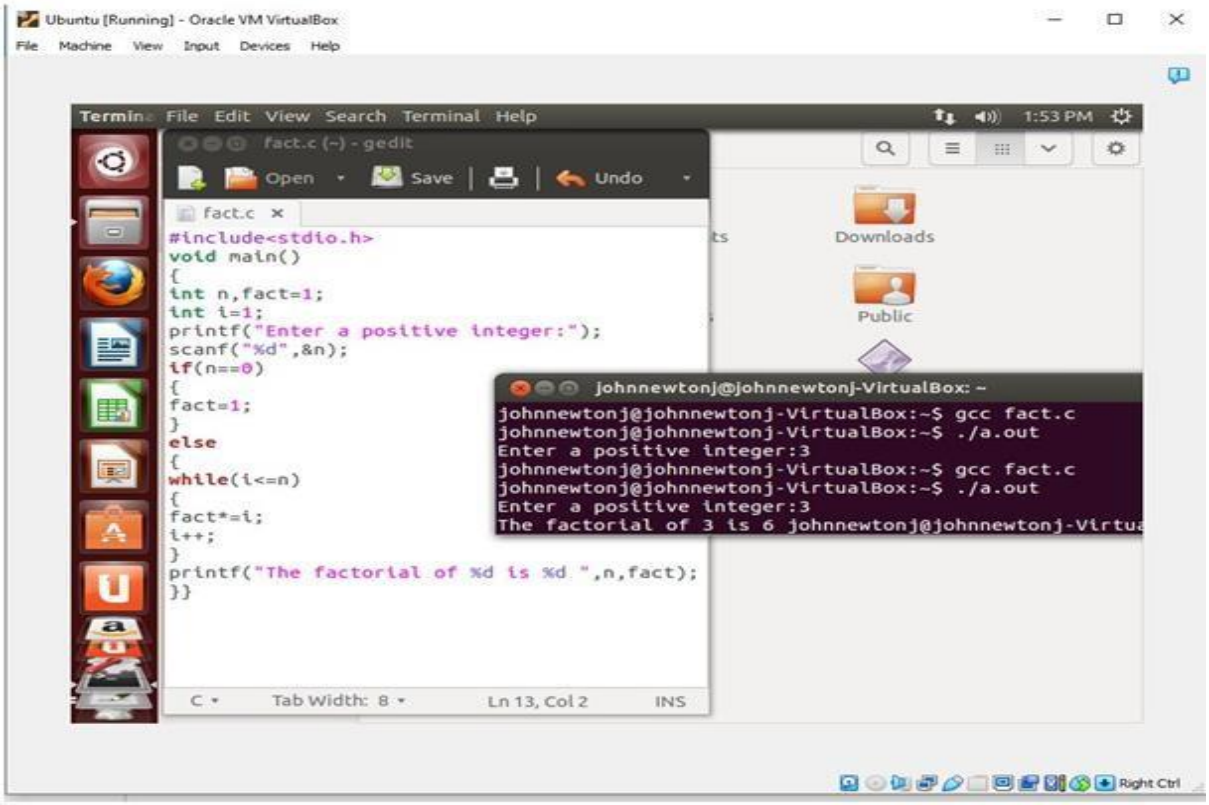
4. Compile and Run the C Program

   cc fact.c

   ./a.out

Output





Result

Thus the simple C programs executed successfully.

Exp. No: 03 **Date:**	**Install Google App Engine and Create a Web Applications using Java**

Aim

Install Google App Engine. Create *hello world* app and other simple web applications using python / Java.

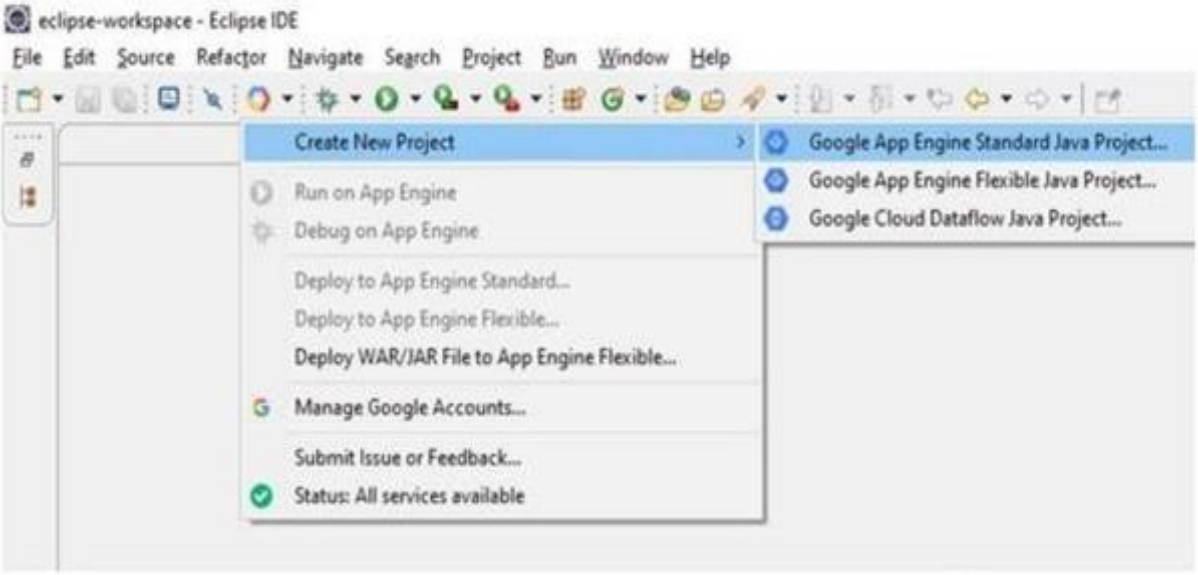
Procedure

Google App Engine SDK Installation

1. Download the Google Cloud SDK installer using the link:
   https://cloud.google.com/appengine/downloads.
2. Select the standard environment as Java.
3. Click Download and Install the Cloud SDK. Launch the installer and follow the prompts.
4. After installation has completed, the installer presents several options:
   Make sure that the following are selected:
   - Start Google Cloud SDKShell.
   - Run **'gcloudinit'**. The installer then starts a terminal window and runs the **gcloudinit** command.
5. Run the following command in your terminal to install the gcloud component that includes the App Engine extension for Java11:
   gcloudcomponentsinstallapp-engine-java

Creating a new App Engine standard project in Eclipse

6. Eclipse with the cloud tools is used to create App Engine standard project.
7. To install the Cloud Tools in Eclipse, select Help > Eclipse Marketplace... and search for "Google Cloud Tools for Eclipse" and click install.
8. After installation restart eclipse when prompted.
9. Click the Google Cloud Platform toolbar button.
10. **Select Create** New Project >Google App Engine Standard Java Project**.**



11. Enter the project name and package name.
12. Click Next. Select the libraries required for the project.
13. Click Finish.
14. The wizard generates a native Eclipse project, with a simple servlet, that you can run and deploy from the IDE.
15. App Engine Java applications use the Java Servlet API to interact with the web server. Modify the defaultHelloAppEngine.java file with your application code.
16. appengine-web.xml is a Google App Engine specific configuration file.
17. web.xml is a standard web application configuration file.
18. Right click the project in the Package Explorer, select **Run As > App Engine**.
19. Eclipse opens its internal web browser to your application.

You can also open an external browser and navigate to http://localhost:8080.

Index.html

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
<head>
<meta http-equiv="content-type" content="application/xhtml+xml; charset=UTF-8" />
<title>Hello App Engine</title>
</head>
<body>
<h1>Hello App Engine!</h1>          <table> <tr>
<td colspan="2" style="font-weight:bold;">Available Servlets:</td>
</tr> <tr>       <td><a href='/hello'>The servlet</a></td>
</tr> </table>    </body>    </html>
```

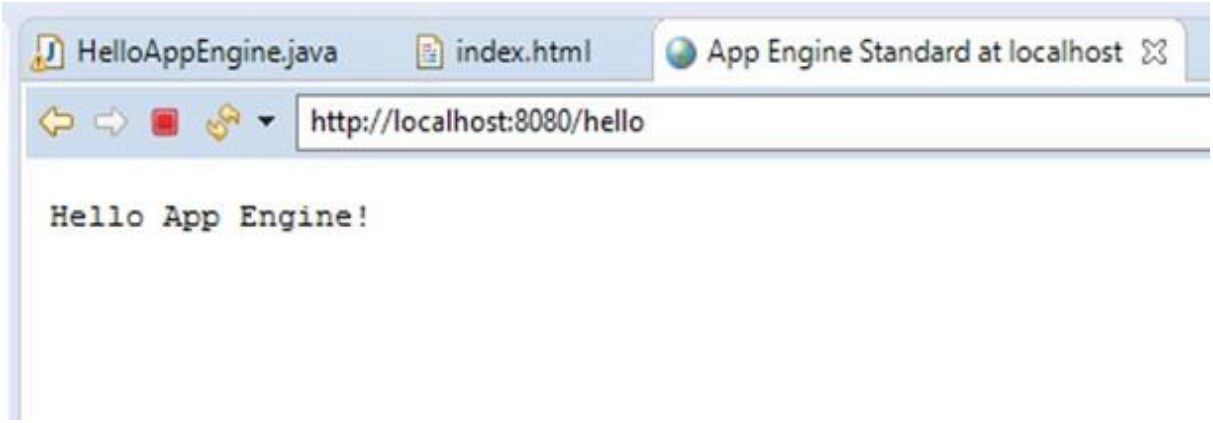
HelloAppEngine.java

```
package com.pack;
import java.io.IOException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet (
        name = "HelloAppEngine",
        urlPatterns = {"/hello"}
)
public class HelloAppEngine extends HttpServlet {
@Override
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws IOException {
        response.setContentType("=text/plain");
        response.setCharacterEncoding("UTF-8");
        response.getWriter().print("Hello App Engine!\r\n");
}       }
```

Input



Output



Result

Thus the simple application was created successfully.

Exp. No: 04 **Date:**	**Use GAE Launcher to Launch the Web Applications**

Aim

Use GAE launcher to launch the web applications.

Procedure

Making your First Application

Now you need to create a simple application. We could use the "+" option to have the launcher make us an application – but instead we will do it by hand to get a better sense of what is going on.

Make a folder for your Google App Engine applications. I am going to make the Folder on my Desktop called "apps" – the path to this folder is:

C:\Documents and Settings\csev\Desktop\apps

And then make a sub--folder in within apps called "ae-01-trivial" – the path to this folder would be:

C:\ Documents and Settings \csev\Desktop\apps\ae-01-trivial

Using a text editor such as JEdit (www.jedit.org), create a file called app.yaml in the ae-01-trivial folder with the following contents:

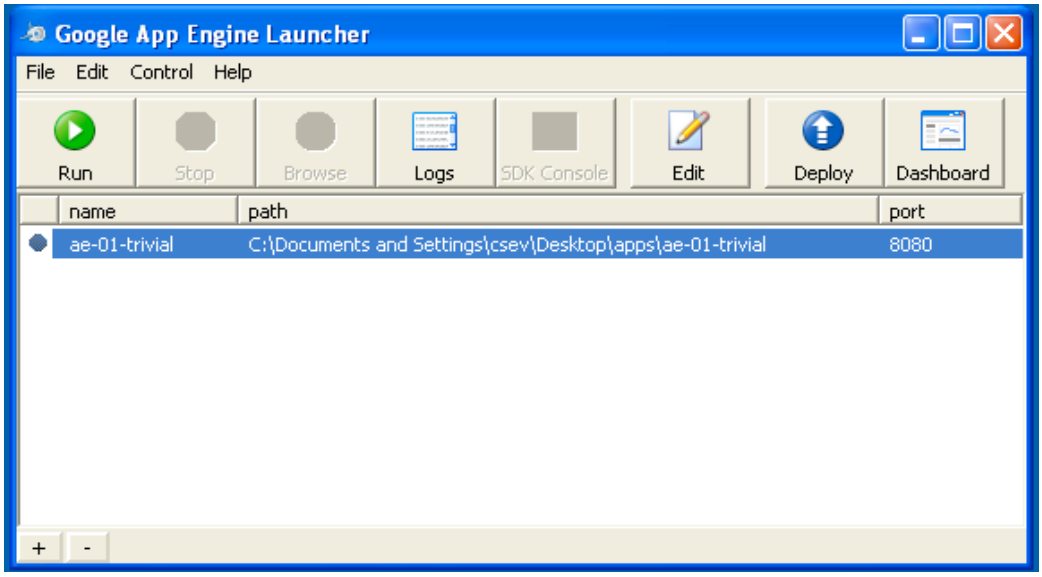
```
application: ae-01-trivial
version: 1
runtime: python api_version: 1
handlers:- url: /.*
script: index.py
```

Note: Please do not copy and paste these lines into your text editor – you might end up with strange characters – simply type them into your editor.

Then create a file in the ae-01-trivial folder called index.py with three lines in it:

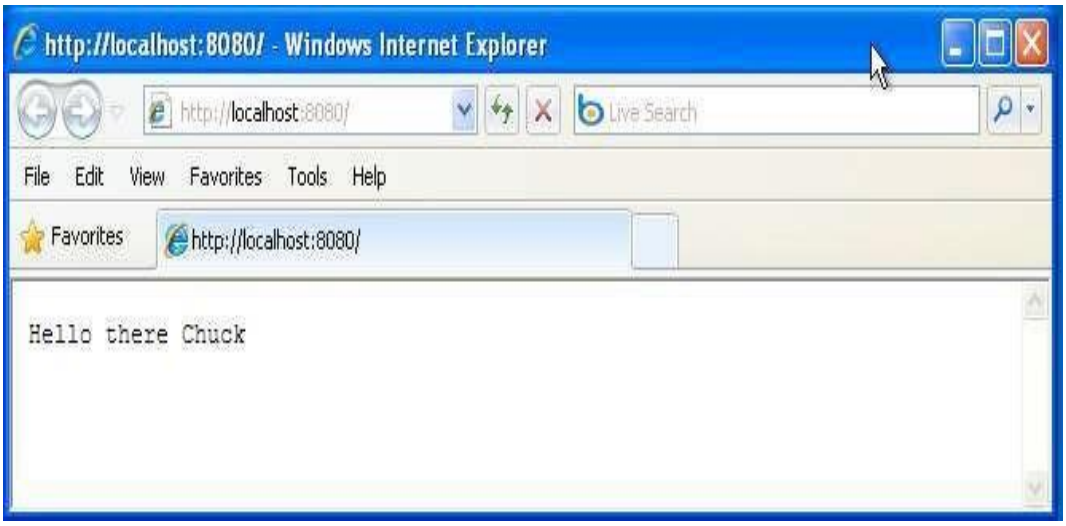
```
print 'Content-Type: text/plain'
print ' '
print 'Hello there Chuck'
```

Then start the GoogleAppEngine Launcher program that can be found under Applications. Use the File -> Add Existing Application command and navigate into the apps directory and select the ae-01-trivial folder. Once you have added the application, select it so that you can control the application using the launcher.



Once you have selected your application and press Run. After a few moments your application will start and the launcher will show a little green icon next to your application. Then press Browse to open a browser pointing at your application which is running at http://localhost:8080/

Paste http://localhost:8080 into your browser and you should see your application as follows:

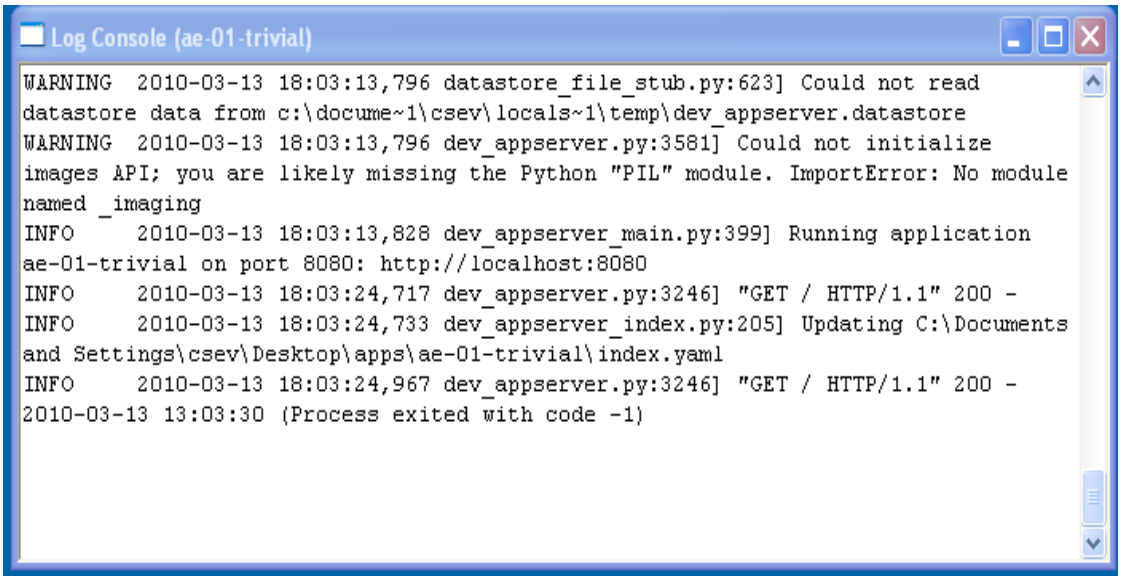


Just for fun, edit the index.py to change the name "Chuck" to your own name and press Refresh in the browser to verify your updates.

Watching the Log

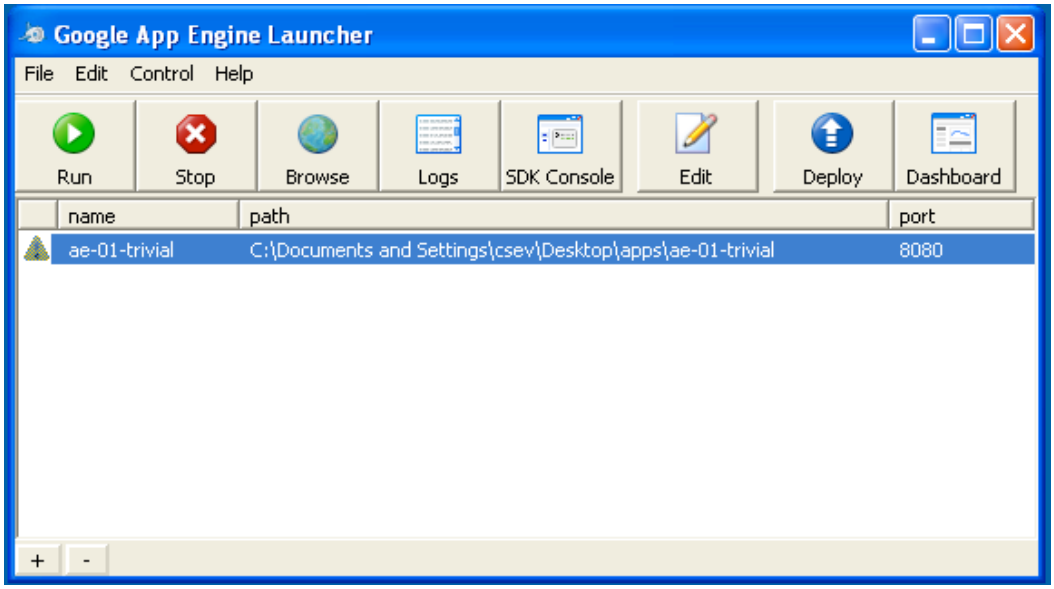
You can watch the internal log of the actions that the web server is performing when you are interacting with your application in the browser. Select your application in the Launcher and press the Logs button to bring up a log window:

Each time you press Refresh in your browser – you can see it retrieving the output with a GET request.



Dealing With Errors

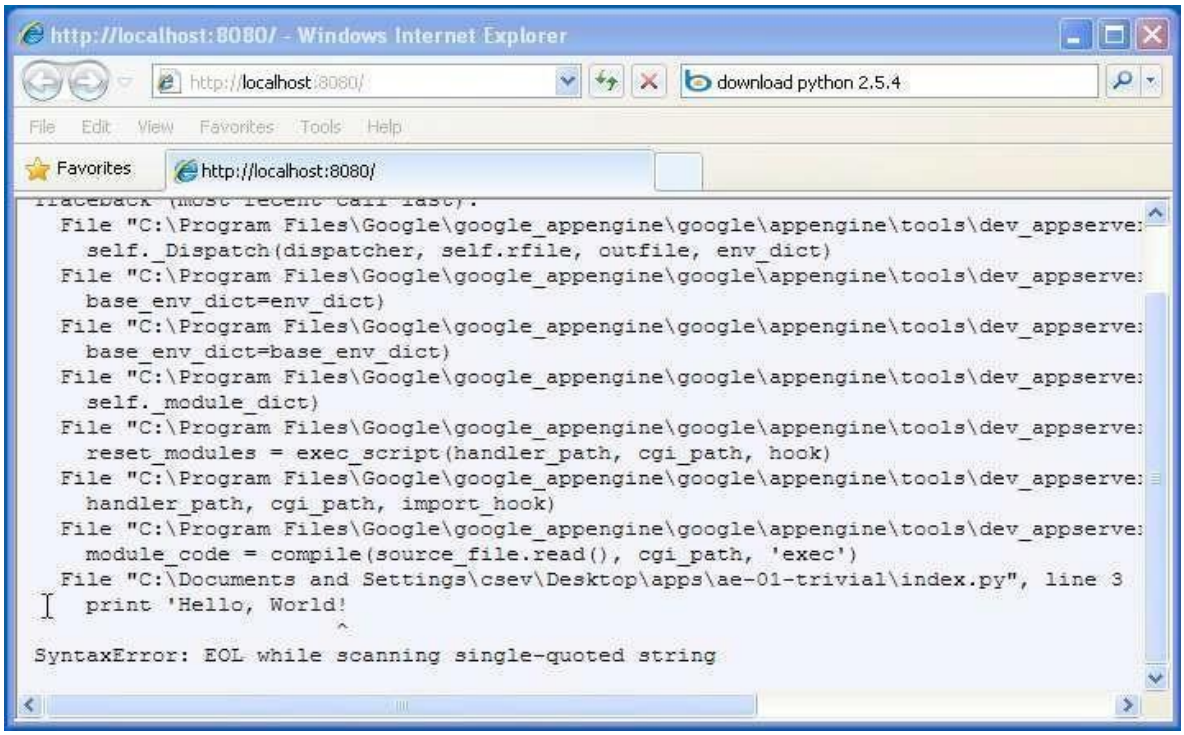
With two files to edit, there are two general categories of errors that you may encounter. If you make a mistake on the app.yaml file, the App Engine will not start and your launcher will show a yellow icon near your application:



To get more detail on what is going wrong, take a look at the log for the application:



In this instance – the mistake is mis--indenting the last line in the app.yaml (line 8). If you make a syntax error in the index.py file, a Python trace back error will appear in your browser.



The error you need to see is likely to be the last few lines of the output – in this case I made a Python syntax error on line one of our one--line application.

Reference: http://en.wikipedia.org/wiki/Stack_trace

When you make a mistake in the app.yaml file – you must the fix the mistake and attempt to start the application again.

If you make a mistake in a file like index.py, you can simply fix the file and press refresh in your browser – there is no need to restart the server.

Shutting Down the Server

To shut down the server, use the Launcher, select your application and press the Stop button.

Result

Thus the GAE web applications was created.

Exp. No: 05 **Date:**	**Simulate a Cloud Scenario Using Cloudsim an Run a Scheduling Algorithm**

Aim

Simulate a cloud scenario using Cloud Sim and run a scheduling algorithm that is not present in Cloud Sim.

Procedure

To use Cloudsim in Eclipse:

1. Download CloudSim installable files
   from *https://code.google.com/p/cloudsim/downloads/list and unzip*
2. Open Eclipse
3. Create a new Java Project: File -> New
4. Import an unpacked CloudSim project into the new Java Project
   The first step is to initialise the CloudSim package by initialising the CloudSim library, as follows:
   CloudSim.init(num_user, calendar, trace_flag)
5. Data centres are the resource providers in CloudSim; hence, creation of data centres is a second step. To create Datacenter, you need the DatacenterCharacteristics object that stores the properties of a data centre such as architecture, OS, list of machines, allocation policy that covers the time or spaceshared, the time zone and its price:
   Datacenter datacenter9883 = new Datacenter(name, characteristics, new VmAllocationPolicySimple(hostList), s
6. The third step is to create a broker:
   DatacenterBroker broker = createBroker();
7. The fourth step is to create one virtual machine unique ID of the VM, userId ID of the VM's owner, mips, number Of Pes amount of CPUs, amount of RAM, amount of bandwidth, amount of storage, virtual machine monitor, and cloudletScheduler policy for cloudlets:
   Vm vm = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new CloudletSchedulerTimeShared())
8. Submit the VM list to the broker:
   broker.submitVmList(vmlist)

9. Create a cloudlet with length, file size, output size, and utilisation model:
   Cloudlet cloudlet = new Cloudlet(id, length, pesNumber, fileSize, outputSize, utilizationModel, utilizationMode
10. Submit the cloudlet list to the broker:
    broker.submitCloudletList(cloudletList)
    Sample Output from the Existing Example:
    Starting CloudSimExample1... Initialising...

Starting CloudSim version 3.0 Datacenter_0 is starting...
>>>

Output

```
========== OUTPUT ==========
Cloudlet ID  STATUS  Data center ID   VM ID   Time   Start Time
    Finish Time 0        SUCCESS   2              0            400
              0.1            400.1
*****Datacenter:
Datacenter_0***** User id
            Debt
   3        35.6

CloudSimExample1 finished!
```

Result

Thus the simple application was created successfully.

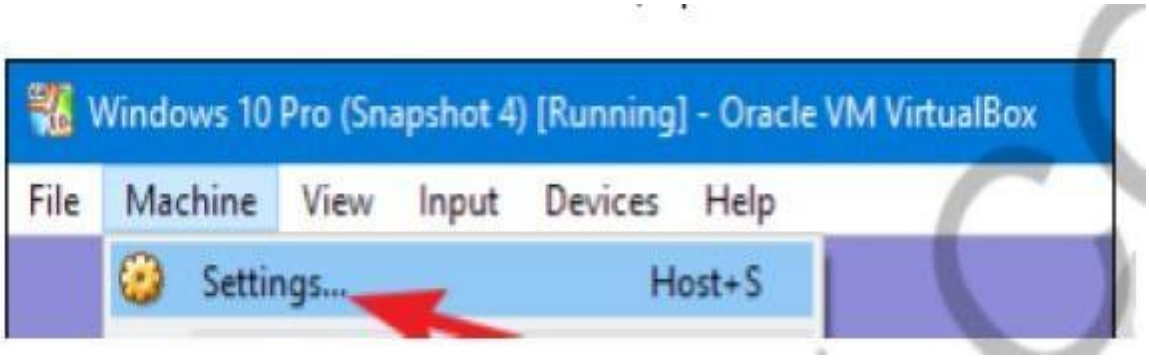
Exp. No: 06 **Date:**	**Copy Files from One Virtual Machine to Another**

Aim

Find a procedure to transfer the files from one virtual machine to another virtual machine.

Procedure

1. Create one shared folder in virtual box.
2. VirtualBox's Shared Folders feature works with both Windows and Linux guest operating systems.
3. To use the feature, you first need to install VirtualBox's Guest Additions in the guest virtual machine.
4. With the virtual machine running, click the "Devices" menu and choose the "Insert Guest Additions CD image" option.
5. This inserts a virtual CD that you can use within the guest operating system to install the Guest Additions.
6. After the Guest Additions are installed, open the "Machine" menu and click the "Settings" option.



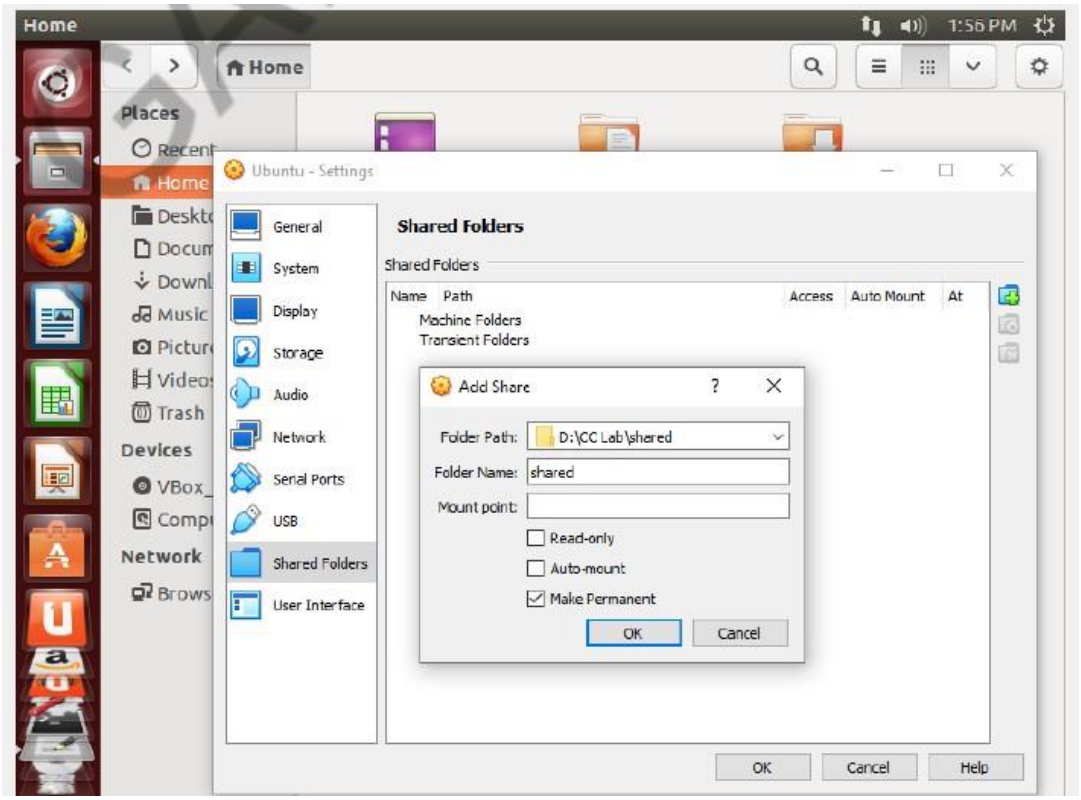
7. In the "Settings" window, switch to the "Shared Folders" tab.
8. Here you can see any shared folders you‟ve set up.
9. There are two types of shared folders.
10. Machine Folders are permanent folders that are shared until you remove them.
11. Transient Folders are temporary and are automatically removed when you restart or shut down the virtual machine.
12. Click the "Add" button (the folder with a plus on it) to create a new shared folder.

In the “Add Share” window, you can specify the following:

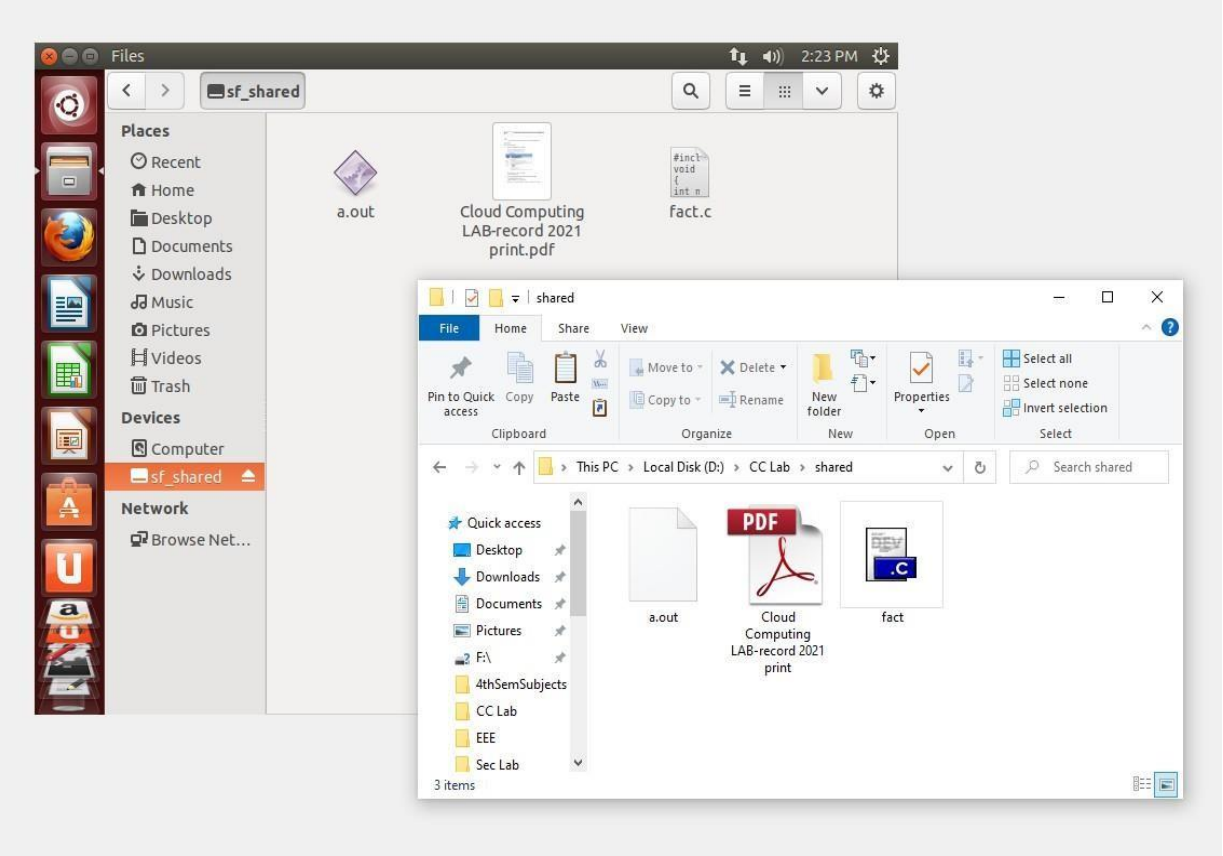
- Folder Path: This is the location of the shared folder on your host operating system (your real PC).
- Folder Name: This is how the shared folder will appear inside the guest operating system.
- Read-only: By default, the virtual machine has full read-write access to the shared folder.
- Enable the “Read-only” checkbox if you want the virtual machine only to be able to read files from the shared folder, but not modify them.
- Auto-mount: This option makes the guest operating system attempt to automatically mount the folder when it boots.
- Make Permanent: This option makes the shared folder a Machine Folder. If you don’t select this option, it becomes a transient folder that is removed with the virtual machine restarts.

Make all your choices and then hit the “OK” button.

Input



Output



Result

Thus the file transfer between VM was successfully completed.

Exp. No: 07 **Date:**	**Create, Deploy and Launch Virtual Machines in Openstack**

Aim

Find a procedure to launch virtual machine using try stack (Online Open stack Demo Version)

Procedure

Step 1: Create Network

Network? Yes, the network in here is our own local network. So, your instances will be not mixed up with the others. You can imagine this as your own LAN (Local Area Network) in the cloud.

1. Go to **Network > Networks** and then click **Create Network**.
2. In **Network** tab, fill **Network Name** for example internal and then click **Next**.
3. In **Subnet** tab,
    1. Fill **Network Address** with appropriate CIDR, for example 192.168.1.0/24. Use private network CIDR block as the best practice.
    2. Select **IP Version** with appropriate IP version, in this case IPv4.
    3. Click **Next**.
4. In **Subnet Details** tab, fill **DNS Name Servers** with 8.8.8.8 (Google DNS) and then click **Create**.

Step 2: Create Instance

Now, we will create an instance. The instance is a virtual machine in the cloud, like AWS EC2. You need the instance to connect to the network that we just created in the previous step.

1. Go to **Compute > Instances** and then click **Launch Instance**.
2. In **Details** tab,
    1. Fill **Instance Name**, for example Ubuntu 1.
    2. Select **Flavor**, for example m1.medium.
    3. Fill **Instance Count** with **1**.
    4. Select **Instance Boot Source** with **Boot from Image**.
    5. Select **Image Name** with **Ubuntu 14.04 amd64 (243.7 MB)** if you want install Ubuntu 14.04 in your virtual machine.

3. In **Access & Security** tab,

1. Click [**+**] button of **Key Pair** to import key pair. This key pair is a public and private key that we will use to connect to the instance from our machine.
2. In **Import Key Pair** dialog,
    1. Fill **Key Pair Name** with your machine name (for example Edward-Key).
    2. Fill **Public Key** with your **SSH public key** (usually is in ~/.ssh/id_rsa.pub). See description in Import Key Pair dialog box for more information. If you are using Windows, you can use **Puttygen** to generate key pair.
    3. Click **Import key pair**.
3. In **Security Groups**, mark/check **default**.
4. In **Networking** tab,
    1. In **Selected Networks**, select network that have been created in Step 1, for example internal.
5. Click **Launch**.
6. If you want to create multiple instances, you can repeat step 1-5. I created one more instance with instance name Ubuntu 2.

Step 3: Create Router

I guess you already know what router is. In the step 1, we created our network, but it is isolated. It doesn't connect to the internet. To make our network has an internet connection, we need a router that running as the gateway to the internet.

1. Go to **Network > Routers** and then click **Create Router**.
2. Fill **Router Name** for example router1 and then click **Create router**.
3. Click on your **router name link**, for example router1, **Router Details** page.
4. Click **Set Gateway** button in upper right:
    1. Select **External networks** with **external**.
    2. Then OK.
5. Click **Add Interface** button.
    1. Select **Subnet** with the network that you have been created in Step 1.
    2. Click **Add interface**.
6. Go to **Network > Network Topology**. You will see the network topology. In the example, there are two network, i.e. external and internal, those are bridged by a router. There are instances those are joined to internal network.

Step 4: Configure Floating IP Address

Floating IP address is public IP address. It makes your instance is accessible from the internet. When you launch your instance, the instance will have a private network IP, but no public IP. In OpenStack, the public Ips is collected in a pool and managed by admin (in our case is TryStack). You need to request a public (floating) IP address to be assigned to your instance.

1. Go to **Compute > Instance**.
2. In one of your instances, click **More > Associate Floating IP**.
3. In **IP Address**, click Plus [**+**].
4. Select **Pool** to **external** and then click **Allocate IP**.
5. Click **Associate**.
6. Now you will get a public IP, e.g. 8.21.28.120, for your instance.

Step 5: Configure Access & Security

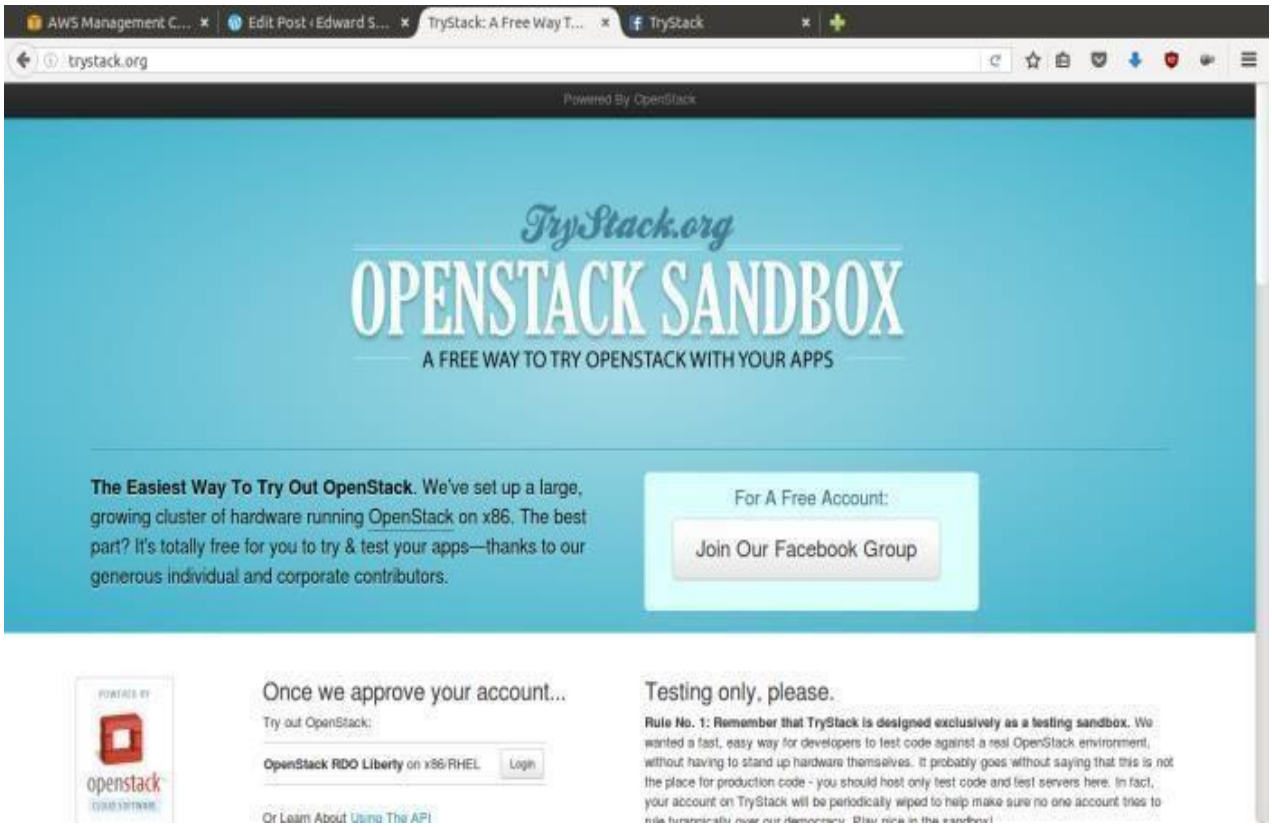
OpenStack has a feature like a firewall. It can whitelist/blacklist your in/out connection. It is called *Security Group*.

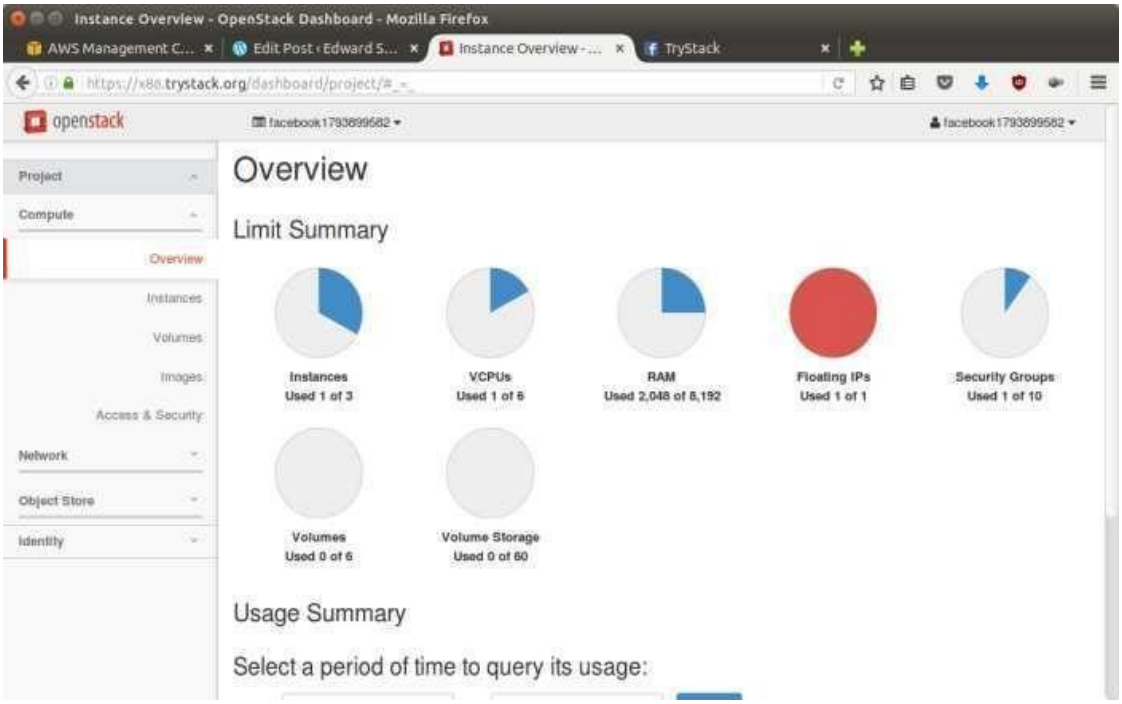
1. Go to **Compute > Access & Security** and then open **Security Groups** tab.
2. In **default** row, click **Manage Rules**.
3. Click **Add Rule**, choose **ALL ICMP** rule to enable ping into your instance, and then click **Add**.
4. Click **Add Rule**, choose **HTTP** rule to open HTTP port (port 80), and then click **Add**.
5. Click **Add Rule**, choose **SSH** rule to open SSH port (port 22), and then click **Add**.
6. You can open other ports by creating new rules.

Step 6: SSH to Your Instance

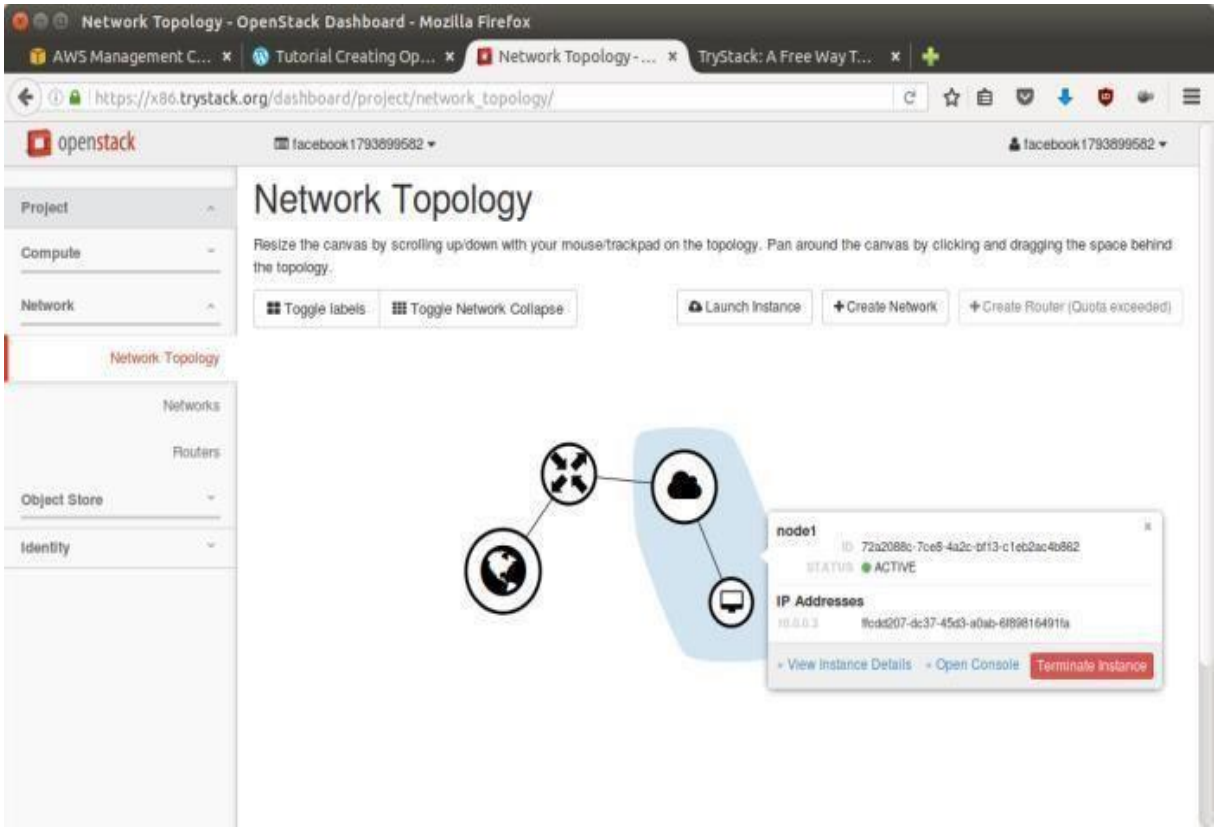
Now, you can SSH your instances to the floating IP address that you got in the step 4. If you are using Ubuntu image, the SSH user will be ubuntu.

Input


AWS Management C...
Edit Post ‹ Edward S...
TryStack: A Free Way T...
TryStack
trystack.org
Powered By OpenStack
TryStack.org
OPENSTACK SANDBOX
A FREE WAY TO TRY OPENSTACK WITH YOUR APPS
The Easiest Way To Try Out OpenStack. We've set up a large, growing cluster of hardware running OpenStack on x86. The best part? It's totally free for you to try & test your apps—thanks to our generous individual and corporate contributors.
For A Free Account:
Join Our Facebook Group
Once we approve your account...
Try out OpenStack:
OpenStack RDO Liberty on x86 RHEL
Login
Or Learn About Using The API
Testing only, please.
Rule No. 1: Remember that TryStack is designed exclusively as a testing sandbox. We wanted a fast, easy way for developers to test code against a real OpenStack environment, without having to stand up hardware themselves. It probably goes without saying that this is not the place for production code - you should host only test code and test servers here. In fact, your account on TryStack will be periodically wiped to help make sure no one account tries to


Instance Overview - OpenStack Dashboard - Mozilla Firefox
AWS Management C...
Edit Post ‹ Edward S...
Instance Overview - ...
TryStack
https://x86.trystack.org/dashboard/project/#_=_
openstack
facebook1793899582
Project
Compute
Overview
Instances
Volumes
Images
Access & Security
Network
Object Store
Identity
Overview
Limit Summary
Instances
Used 1 of 3
VCPUs
Used 1 of 6
RAM
Used 2,048 of 8,192
Floating IPs
Used 1 of 1
Security Groups
Used 1 of 10
Volumes
Used 0 of 6
Volume Storage
Used 0 of 60
Usage Summary
Select a period of time to query its usage:

Output



Result

Thus the openstack demo worked successfully.

Exp. No: 08 **Date:**	**Install Hadoop Single Node Cluster and Run Simple Applications like Word Count**

Aim

Install Hadoop single node cluster and run simple applications like word count.

Procedure

1. Download Hadoop 2.8.0
2. Check either Java 1.8.0 is already installed on your system or not, use “Javac - version" to check Java version
3. If Java is not installed on your system then first install java under "C:\JAVA" Java setup
4. Extract files Hadoop 2.8.0.tar.gz or Hadoop-2.8.0.zip and place under "C:\Hadoop-2.8.0" hadoop
5. Set the path HADOOP_HOME Environment variable.
6. Set the path JAVA_HOME Environment variable.
7. Next we set the Hadoop bin directory path and JAVA bin directory path.

Hadoop Configuration with Program

1. File C:/Hadoop-2.8.0/etc/hadoop/core-site.xml, paste below xml paragraph and save this file.

```
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

2. Rename mapred-site.xml.template" to "mapred-site.xml" and edit this file

C:/Hadoop2.8.0/etc/hadoop/mapred-site.xml, paste below xml paragraph and save this file.

```
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
```

3. Create folder "data" under "C:\Hadoop-2.8.0"
   - Create folder "datanode" under "C:\Hadoop-2.8.0\data"
   - Create folder "namenode" under "C:\Hadoop-2.8.0\data" data

4. Edit file C:\Hadoop-2.8.0/etc/hadoop/hdfs-site.xml, paste below xml paragraph and save

this file.

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>C:\hadoop-2.8.0\data\namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>C:\hadoop-2.8.0\data\datanode</value>
</property>
</configuration>
```

5. Edit file C:/Hadoop-2.8.0/etc/hadoop/yarn-site.xml, paste below xml paragraph and save this file.

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
```

6. Edit file C:/Hadoop-2.8.0/etc/hadoop/yarn-site.xml, paste below xml paragraph and save this file.

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
```

7. Edit file C:/Hadoop-2.8.0/etc/hadoop/hadoop-env.cmd by closing the command line "JAVA_HOME=%JAVA_HOME%" instead of set "JAVA_HOME=C:\Java"

Hadoop Configuration

1. Download file Hadoop Configuration.zip
2. Delete file bin on C:\Hadoop-2.8.0\bin, replaced by file bin on file just download (from Hadoop Configuration.zip).
3. Open cmd and typing command "hdfs namenode –format". You will see hdfs namenode –format Testing.
4. Open cmd and change directory to "C:\Hadoop-2.8.0\sbin" and type "start- all.cmd" to start apache.
5. Make sure these apps are running.
   - Name node
   - Hadoop data node
   - YARN Resource Manager
   - YARN Node Manager hadoop nodes .
6. Open: http://localhost:8088.

Open cmd in Administrative mode and move to "C:/Hadoop-2.8.0/sbin" and start cluster

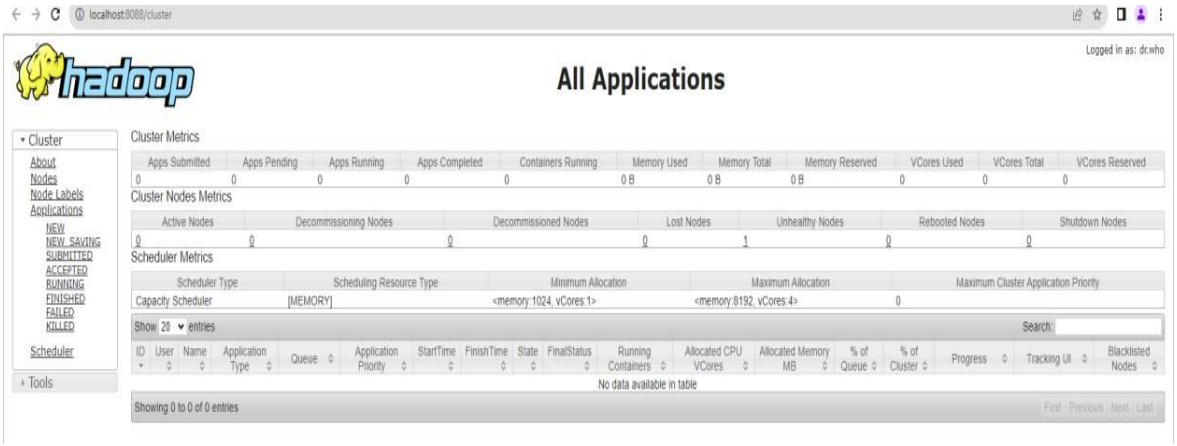
1. Create an input directory in HDFS.
   hadoop fs –mkdir/input_dir
2. Copy the input text file named input_file.txt in the input directory (input_dir) of HDFS.
   hadoop fs –put C:/input_file.txt/input_dir
3. Verify input_file.txt available in HDFS input directory (input_dir).
   hadoop fs –ls/input_dir/
4. Verify content of the copied file.
   hadoop dfs –cat/input_dir/input_file.txt
5. Run MapReduceClient.jar and also provide input and out directories.
   hadoop jar C:/MapReduceClient.jar wordcount /input_dir /output_dir
6. Verify content for generated output file.
   hadoop dfs -cat /output_dir/*

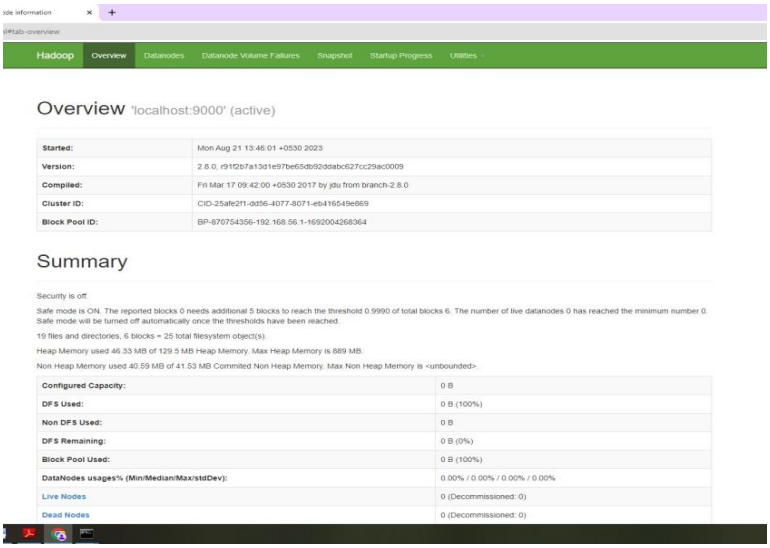
Input

```
C:\hadoop-2.8.0\sbin>hadoop dfs -cat /output_dir/*
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
23      12
24      6
25      18
26      36
27      12
28      24
29      6
30      24
31      24
32      18
33      6
34      30
35      6
36      12
38      24
39      66
40      18
41      24
42      6
43      12
45      6
cloud   2
manickam        2

C:\hadoop-2.8.0\sbin>
```

Output





Result

Thus the Hadoop one cluster was installed and simple applications executed successfully.

Exp. No: 09 **Date:**	**Create and Use a Repository using Git Hub**

Aim

Create and use a repository using git hub

Procedure

Create and Use a GitHub Repository

Prerequisites

- Git installed on your system
- GitHub account created
- Basic terminal or command prompt access

Step 1: Create a New Repository on GitHub

1. **Log in to GitHub**: Go to https://github.com and sign in.
2. **Click "New"**: On the top-right corner, click the "+" icon and select **New repository**.
3. **Fill in details**:
    - Repository name: cloud-lab
    - Description: *Optional*
    - Choose visibility: Public or Private
    - Initialize with README: ✓(recommended)
4. **Click "Create repository"**

Step 2: Clone the Repository to Your Local Machine

1. **Copy the HTTPS URL**: Click the green **Code** button and copy the HTTPS link.
2. **Open Terminal**:
   git clone https://github.com/your-username/cloud-lab.git
3. **Navigate into the folder**:
   cd cloud-lab

Step 3: Add Files to the Repository

1. **Create a file**:
   echo "# Cloud Computing Lab" > lab1.txt
2. **Check status**:
   git status
3. **Stage the file**:
   git add lab1.txt
4. **Commit the changes**:
   git commit -m "Added lab1.txt for cloud computing lab"

Step 4: Push Changes to GitHub

1. **Push to remote**:
   git push origin main

Expected Output Screen:

- Terminal shows:
  To https://github.com/your-username/cloud-lab.git

2. **Verify on GitHub**:
   - Go to your GitHub repo page.
   - You'll see lab1.txt listed under the files.

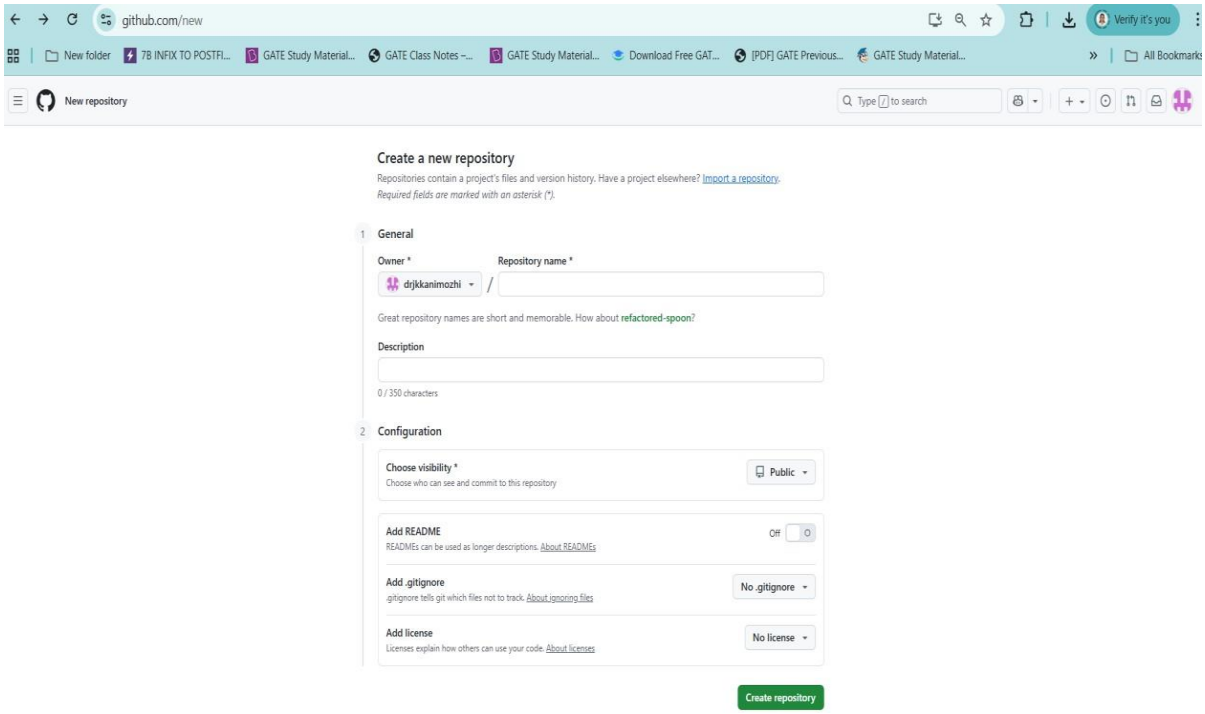
Step 5: Modify and Update Files

1. **Edit the file**:
   echo "This is Lab 1 content" >> lab1.txt
2. **Repeat add, commit and push**:
   git add lab1.txt
   git commit -m "Updated lab1.txt with lab content"
   git push origin main

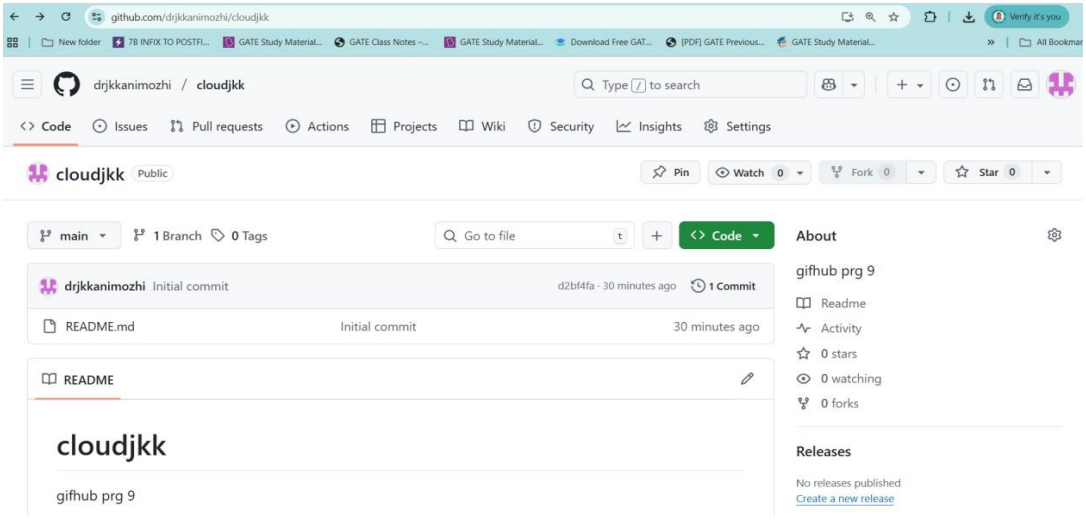
Summary of Commands

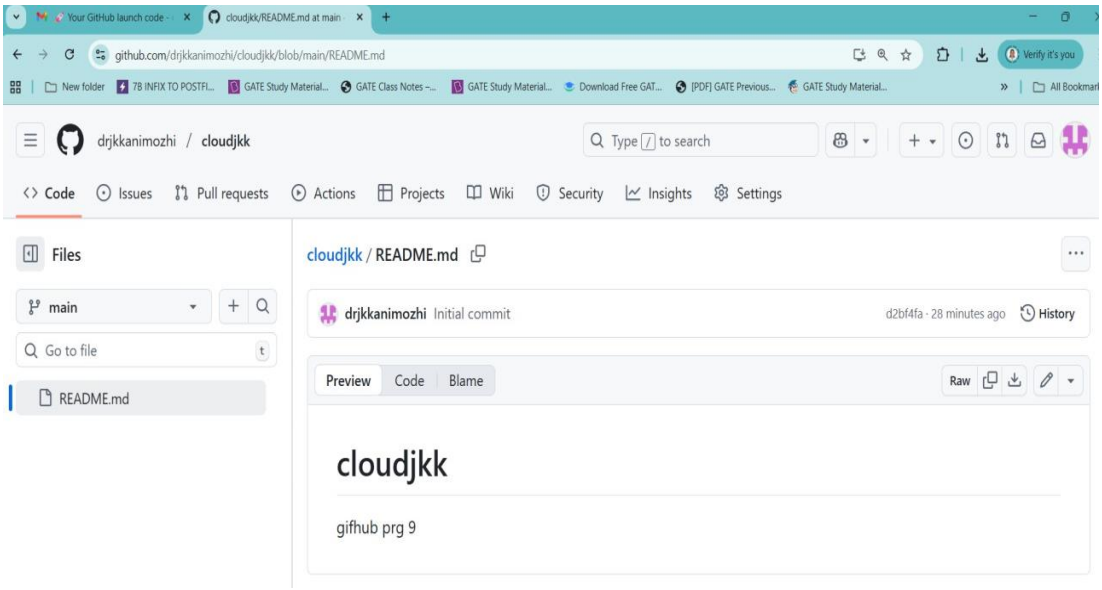
git clone <repo-url>
cd <repo-folder>
git add <filename>
git commit -m "message"
git push origin main

Input



Output





Result

Thus the simple application was created successfully.

Exp. No: 10 Date:	Create a Blog for MCA Department Profile

Aim

Create a blog to show the profile of our MCA department

Procedure

Step 1: Choose a Blogging Platform

Recommended Options:

- Blogger
- WordPress
- GitHub Pages (for tech-savvy users)

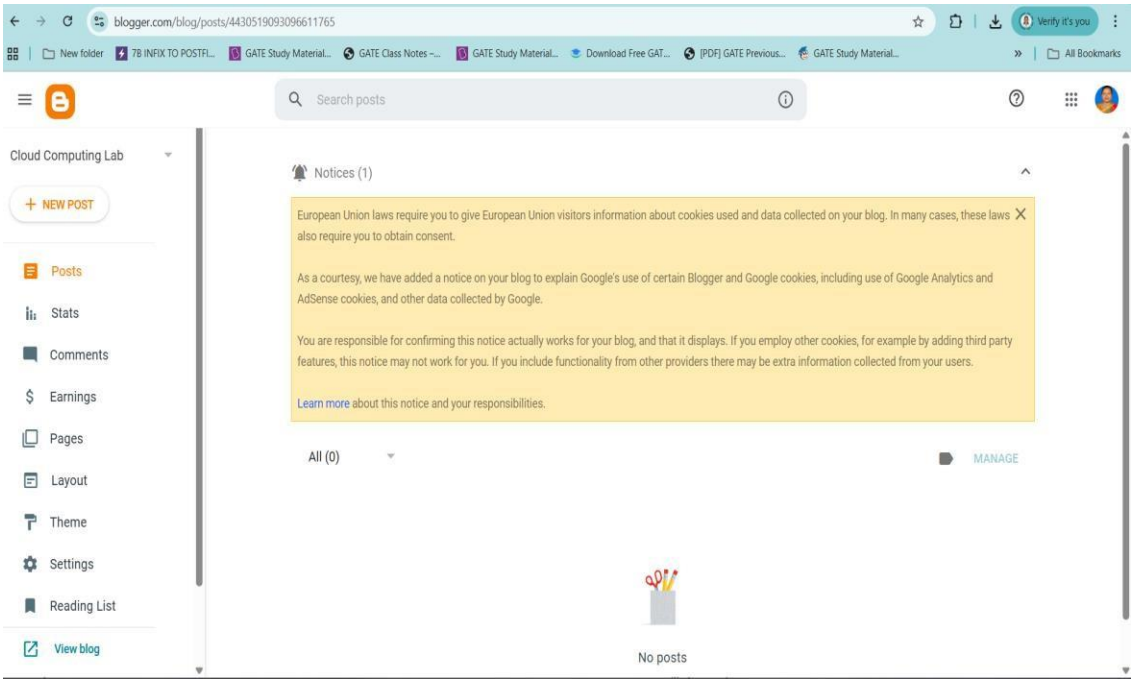
For simplicity, we'll use **Blogger**.

Step 2: Sign In to Blogger

Procedure

1. Go to https://www.blogger.com
2. Sign in using your Google account.

Output Design



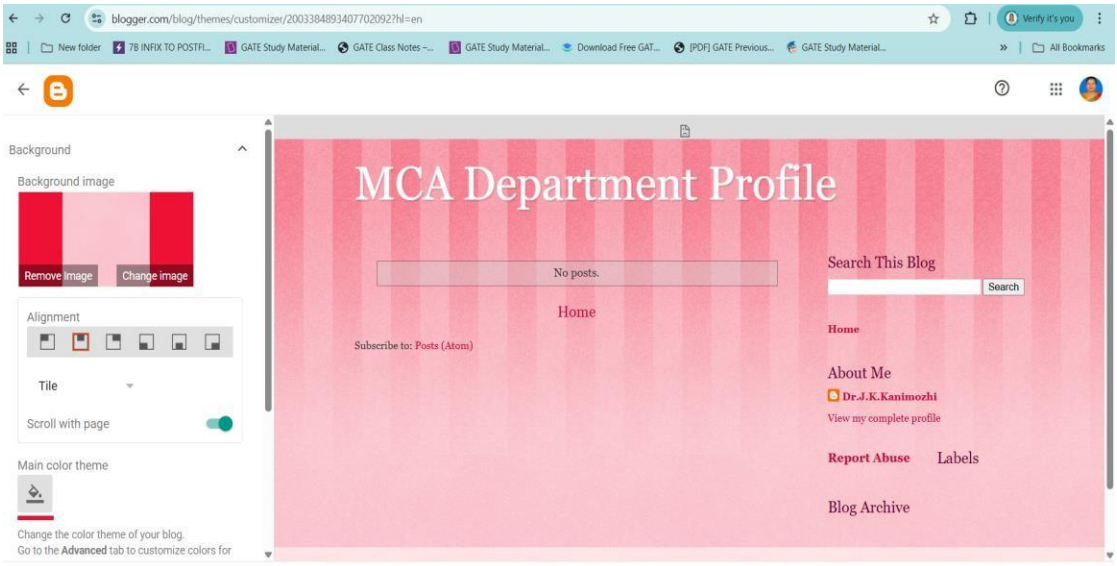
Welcome to Blogger!
[Create New Blog] button visible

Step 3: Create a New Blog

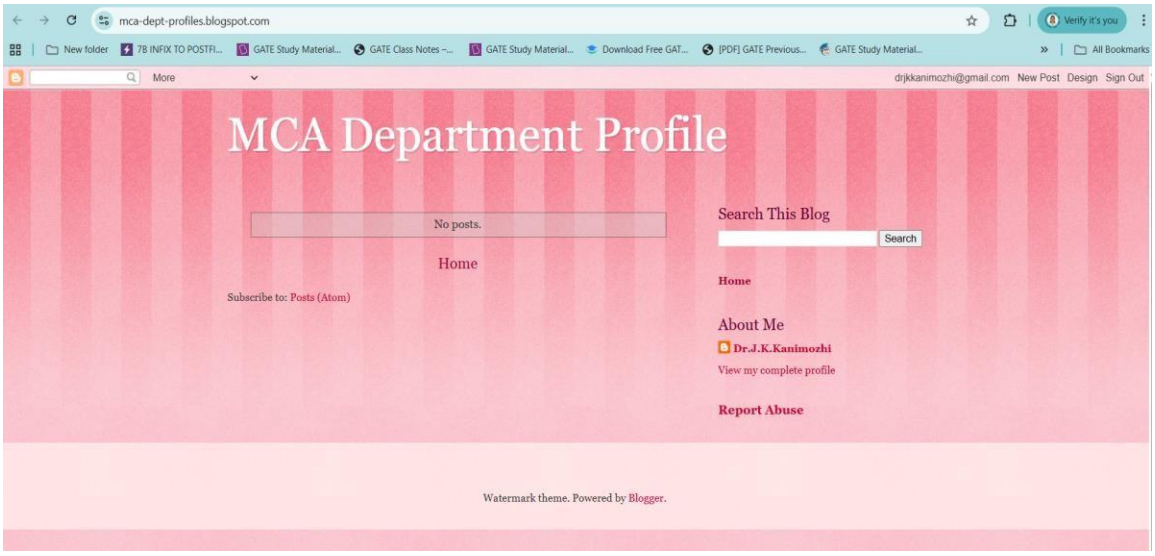
Procedure

1. Click **Create New Blog**
2. Fill in:
    - **Title**: MCA Department Profile
    - **Address**: mca-dept-profile.blogspot.com (or any available name)
    - **Theme**: Choose a clean, professional theme
3. Click **Create Blog**

Output Design



Blog Created: MCA Department Profile
URL: https://mca-dept-profile.blogspot.com
Theme: Contempo (or your choice)



Step 4: Create a New Post

Procedure

1. Click **New Post**
2. Title: *About MCA Department*
3. Content:
   MCA Department Profile

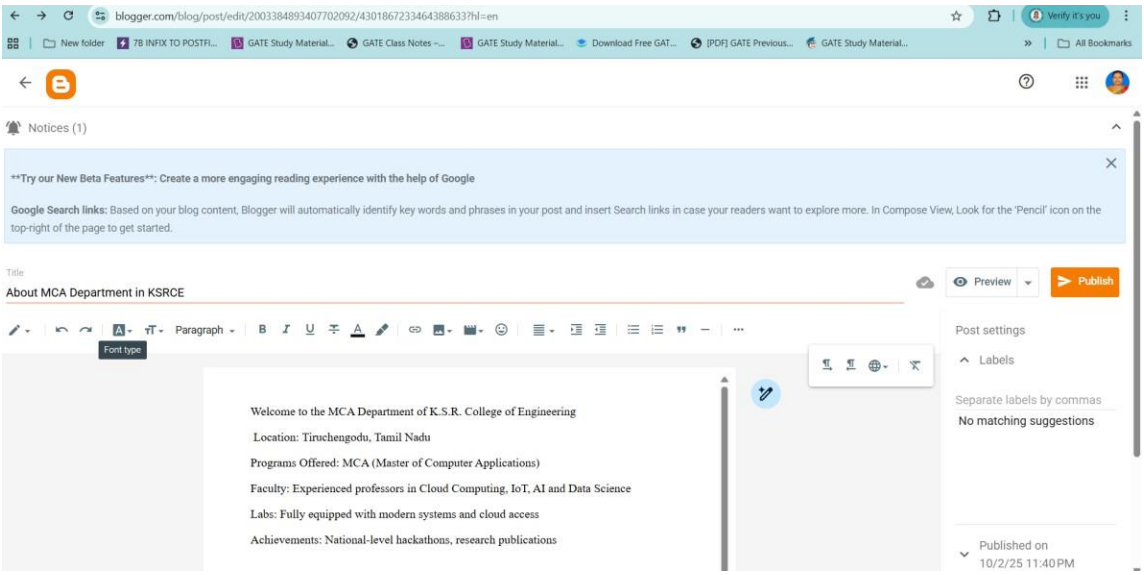
   Welcome to the MCA Department of [Your College Name].

   Location: Tiruchengodu, Tamil Nadu
   Programs Offered: MCA (Master of Computer Applications)
   Faculty: Experienced professors in Cloud Computing, IoT, AI and Data Science
   Labs: Fully equipped with modern systems and cloud access
   Achievements: National-level hackathons, research publications

The Department of Computer Applications (MCA) was started in the year 2003. Started with an intake of 60 students, the sanctioned intake was increased to 1 20 seats in 2007. In the year 2010, the sanctioned intake was increased to 180 seats.
The department has organized various workshops in collaboration with the industry to keep abreast with the latest updates and conducted lecture series by eminent personalities from the industry and academia. To foster creativity amongst the students and faculty it brings forth National Conference titled Emerging Issues in Computer Applications once in a year.
To enlighten and explore every mind technically, a National Level Technical Symposium "Zelos" is being organized by our Students on a yearly basis. Thus, the post-graduation program produces ideal Software Professionals, well equipped to meet the challenges of the rapidly growing IT industry.

4. Add images (department building, lab, events)
5. Click **Publish**

Output Design



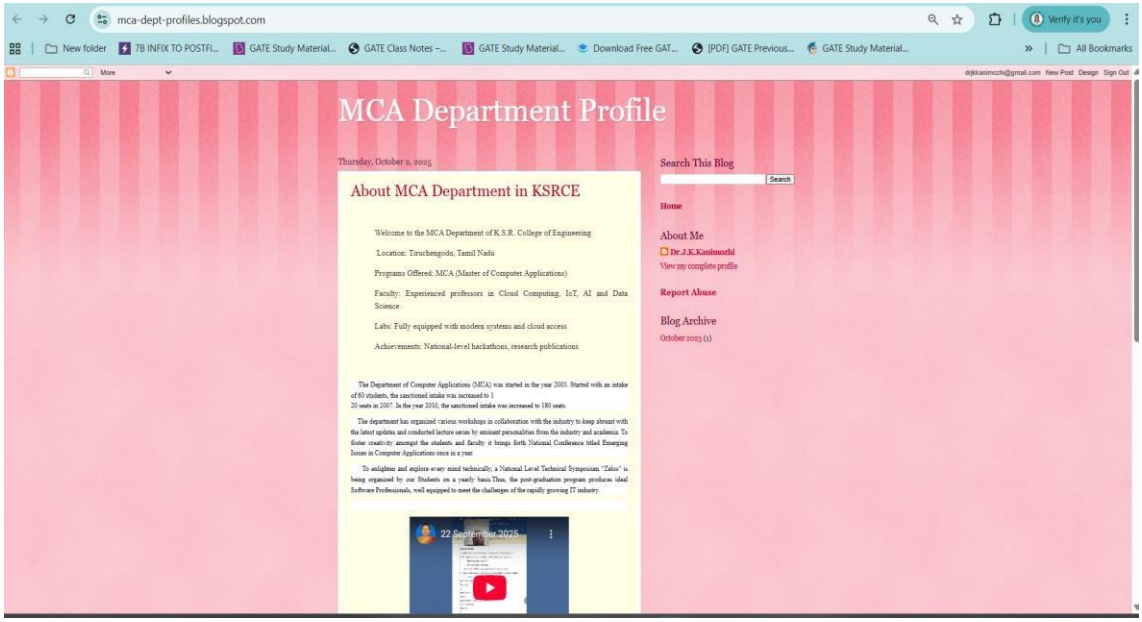
Post Title: About MCA Department
Status: Published

Step 5: View and Share the Blog

Procedure

1. Click **View Blog**
2. Copy the blog URL (https://mca-dept-profiles.blogspot.com/)
3. Share with classmates, faculty, or embed in college website

Output Design



Blog URL: https://mca-dept-profile.blogspot.com
Post: About MCA Department
Content: Displays profile, images and achievements

Final Blog Structure

MCA Department Profile
├── About MCA Department (Post)
│ ├── Description
│ ├── Faculty Info
│ ├── Lab Details
│ └── Achievements
└── Images: Department, Events, Labs

Result

Thus the simple application was created successfully.